ESSENTIAL - STELLAR CYBER OPEN XDR ARCHITECTURE OVERVIEW TRANSCRIPT

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L00 - OPEN XDR ARCHITECTURE - INTRODUCTION AND AGENDA

L00A-PRES: Introduction and Agenda (02m:42s)

00:00-01:00
Hi. Welcome to the *Open XDR Architecture Overview* module. I am Paul Levasseur of the Stellar Cyber Enablement and Training teams. The lessons in this module provide an overview of the major components and features within the Stellar Cyber Open XDR architecture. This module is a foundation for the modules and lessons in the Stellar Cyber Open XDR training curriculum. Throughout this module, there'll be lots of new concepts and terms that you may not be familiar with, but that's okay and to be expected. The important thing is that you become familiar with the concepts and terms described. As you progress through the Stellar Cyber Open XDR training curriculum, with repetition, iteration, and examples you'll be able to understand and apply what you have learned in becoming a cybersecurity superstar. Thank you.

01:00-02:00
In the *Stellar Cyber Open XDR Architecture Overview* module, we will cover the following lessons:

- *The Big Picture*. This lesson is based on a diagram showing all the major components of the Stellar Cyber Open XDR architecture, along with how the components interact and communicate with each other.

- *Open XDR Platform*. This lesson provides an overview of the major components and functionality of the Stellar Cyber Open XDR platform.

- *Modular Sensors*. This lesson describes the components and features of Stellar Cyber Modular Sensors, which are essential for data collection from security, identity, infrastructure tools, and network traffic, and for response actions to mitigate threats.

- *Server Sensors*. This lesson provides an overview of how Windows Server sensors and Linux Server sensors collect and forward data to the Stellar Cyber Open XDR platform, and

02:00-02:41

- *Data Collection Overview*. This lesson focuses on how the Stellar Cyber platform collects and ingests data from all the security, infrastructure, and identity tools within an organization to provide threat detection, investigation, and response actions to detect and mitigate threats.

The lessons in this module provide an overview of the major components and features within the Stellar Cyber Open XDR architecture. This module is the foundation for the modules and lessons in the Stellar Cyber Open XDR training curriculum.

L01 - THE BIG PICTURE

L01A-PRES: Stellar Cyber Open XDR Architecture Overview-The Big Picture (04m:28s)

00:00-01:00
The *Stellar Cyber Open XDR Architecture Overview* module will start with the lesson, *The Big Picture. The Big Picture*. This lesson is based on a diagram showing all the major components of the Stellar Cyber Open XDR architecture, along with how the components interact and communicate with each other.

The big picture is a diagram representing all the major components of the Stellar Cyber Open XDR architecture. Starting on the right is the Stellar Cyber Open XDR platform. I will go through the main components and functionality of the Stellar Cyber Open XDR platform, including the Data Analyzer, The Data Lake, how Interflow works, Normalization, Enrichment...

01:00-02:00

...machine learning with both supervised ML and unsupervised ML. I will cover the statistical analysis, heuristic rules, and ATH playbooks, used to detect threats. I will cover correlation with Graph ML. I will go into search, the user interface, and the API. On top of that, I will give an overview of our connectors for the cloud, for data collection and response actions to cloud security, services, and apps. I will talk about the data sync, which allows the platform to sync with cold storage, or Data Lake, or other SIEMs. This includes Amazon S3, Azure Blob, Google Cloud Storage, OCI archive Splunk, and Elasticsearch.

02:00-03:00

I will then cover the Stellar Cyber Modular Sensor, which is a major component of the Stellar Cyber architecture providing NDR services, which is network detection and response. I will cover connections from the Modular Sensor to the Stellar Cyber Open XDR platform. This includes the Configuration Management connection, and then the other connection is used to send the data. I will talk about how logs are forwarded to the Modular Sensor from infrastructure and security tools using Syslog in CEF or LEEF format, JSON, NetFlow, and IPFIX. I'll describe how network traffic is sent from a switch to the Modular Sensor, using a SPAN port or TAP device. I'll talk about connectors that are software modules on the Modular Sensor...

03:00-04:00

...used to collect data from local security services and apps. I will provide an overview of how the connectors on Modular Sensors collect information from local security services and apps including Active Directory, firewalls, including web application firewalls and vulnerability scanning services such as Nessus or Qualys. And I'll provide an overview of the Modular Sensor features including log forwarder, including network traffic app identification, session state, process correlation and how metadata is collected. I'll mention how data can be filtered. I'll provide an overview of the IDS functionality, the Malware Sandbox functionality, as well as other components such as the aggregator, the Tenable Nessus scanner, connectors, and how data can be buffered.

04:00-04:28

I will keep on returning to the big picture throughout this lesson so we can see how the components fit within the Stellar Cyber architecture.

The lesson, *The Big Picture* is complete. You should now be familiar with the diagram showing all the major components of the Stellar Cyber Open XDR architecture along with how the components interact and communicate with each other. Thank you.

L02 - STELLAR CYBER OPEN XDR PLATFORM

L02A-PRES: Stellar Cyber Open XDR Platform (22m:02s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Architecture Overview* module is the *Open XDR Platform*. This lesson, *Open XDR Platform*, provides an overview of the major components and functionality of the Stellar Cyber Open XDR platform.

By looking at the big picture, we can see all the components in the Stellar Cyber Open XDR architecture. In this section, I'll focus specifically on the components and functionality of the Stellar Cyber Open XDR platform, as seen on the right side of the picture. To start, I will go through each of the functions of the Stellar Cyber Open XDR platform. Starting with number one, the Data Analyzer. The Data Analyzer is one of the main functional components of the Stellar Cyber platform.

01:00-02:00
That includes receivers, used for data ingestion from sensors and connectors. It does Normalization and Enrichment of data collected within Interflow records. Processing and analyzing security threat intel information such as reputation, and storing data as Interflow records in the Data Lake.

Number two is the Data Lake, or DL. The Data Lake is the central repository for data storage in the Stellar Cyber platform. The Data Lake aggregates and stores data from all sources, enabling the correlation of threats across all data inputs. It also provides the foundation for data analysis and correlation.

02:00-03:00
Number three, Interflow. Interflow is the record format expressed as JSON objects used by Stellar Cyber to represent raw data, events, and anomalies collected from all security identity and infrastructure tools as well as from network traffic within an organization by sensors and connectors. Here's an example excerpt of an Interflow record. Here you'll see there are field names such as xdr underscore event and display underscore name, as well as field values such as internal ransomware. This format is easy for humans and machines alike to read and understand.

In Stellar Cyber, the destination IP field and value from all sources is Normalized to DST IP as a field name...

05:00-06:00
...and then the field value is the IP address. Using Normalized field names and values, Stellar Cyber can apply machine learning and correlate field names and values across all different data sources to identify threats and attacks. SOC analysts can perform consistent searches knowing that searching for DST IP will get you the destination IP across all different data sources used, or if you're creating custom rules using ATH playbooks. Again, you only need to know the field name DST IP. This applies across all field names and values, not just of course IP addresses.

Number five, Enrichment. Enrichment is the process of augmenting Interflow records with additional information from external sources that enhance the value and context of the Interflow records.

06:00-07:00
Fields are added such as IP reputation from threat intel. This will let me know if the IP address is coming from a command and control. It'll let me know if the IP address is coming from a known emerging threat, or it's a Tor node, or it has a good reputation. We add geolocation for IP addresses. This way we know the country, the city, and the location of IP addresses that are received into the platform. The Stellar Cyber platform performs DNS lookups of IP addresses and adds those to the Interflow records. If you have an identity platform connected, for example Active Directory, the username is also added to the Interflow record. Another example is we add IP type.

07:00-08:00
The platform will let you know if the IP is coming from the public internet or it's a private IP address from RFC 1918, which includes the 10 network or 17216 to 17231 or 192168. If I'm creating a rule for example with this information, I can say I want to see all my public to private IP address communication. We also look for multicast, which is two two four two two three nine. Network range. Enrichment is very beneficial and gives you much more context in the Interflow records.

Number six is machine learning. Stellar Cyber uses supervised and unsupervised machine learning models to detect possible threats, reduce false positives and identify security breaches.

08:00–09:00
Next I'll provide examples of supervised machine learning and unsupervised machine learning.

Number seven, supervised machine learning. Supervised machine learning is used to detect known bad patterns based on training of the machine learning using large-scale data sets to define what bad and good look like. Supervised machine learning is used when there are clear identifiers for what bad looks like, but it's far too computationally complex for a simple rule to be used. An example of supervised machine learning is a domain-generation algorithm Alert. Stellar Cyber trains the machine learning with a large number of real domains and bad or malicious domains that are generated typically by domain-generation algorithms, such as x h t z l e q y b dot com...

09:00-10:00

...and a a b b c c d d one one two two three three dot net. You'll see many of these fake domains and Stellar Cyber learns what the fake ones look like, trains the machine learning. And this way also applies some other computational analysis to determine whether or not the domain is good or bad. Stellar Cyber will also determine if host names are resolvable as part of the calculation as well. The Alert is triggered when a host is determined to be using a domain-generation algorithm, which indicates that the host might be compromised.

Number eight, unsupervised machine learning. Unsupervised machine learning uses the concept of non standard patterns to identify bad events that cannot be easily detected by a rule or known bad pattern. Unsupervised machine learning learns the normal pattern within an environment...

10:00-11:00

...and then triggers Alerts when activity is anomalous or deviates significantly from a learned baseline within the environment. Here's an example of unsupervised machine learning. One might be User Login Location Anomaly. This Alert requires at least twenty past logins per user in a time span of fourteen days, beginning from their first login. There's no rule to detect this but the fact that the machine learning, over time, learns what the normal behavior is the Alert triggers when there's a significant deviation from the observed normal pattern of user login locations. And again this is per user.

Number nine, statistical analysis rules.

11:00-12:00

Statistical analysis rules detect threats by analyzing data using statistical methods to identify patterns, trends, or anomalies in the data. An example of a statistical analysis rule is Stellar Cyber uses statistical analysis on Microsoft 365 user deletion events to calculate and compare statistics related to the number of users deleted in a short period of time. If the number is too high, an Alert is triggered. So unlike unsupervised machine learning, which

learns the standard behavior a statistical analysis rule can be based on an average for example. If the average over time deviates by a certain threshold, then trigger the Alert.

Next is a heuristic rule. Heuristic rules detect threats by leveraging rules based on simple arithmetic or logical conditions.

12:00-13:00

An example of this is a bad reputation Alert. The rule looks at simple logic where if the threat intel feed shows an IP address is sourced with a bad reputation, then an Alert is triggered. It's pretty simple. Heuristic rules are more specific and look for specific cases of when things do not match a certain pattern. The Sigma rules built into the system are based on heuristic rules. Next, analysts on the platform can create automated threat hunting playbooks, or ATH playbooks. These are user-defined automation rules with a trigger condition in action. For example, you can have a trigger condition to look for a certain query. Then based on finding something in a query, I can then perform an action. And the action can be used to send an email notification, a Slack notification...

13:00-14:00

...create a webhook to an external server, create an Alert, create a firewall or security switch rule to block an IP address, disable a user in Active Directory, contain an endpoint with an EDR solution, or run a script as an example. Looking at an ATH playbook, you'll see that there's a configuration for a usually a type of query. There's an Alert configuration then I define a condition, then I perform the action. There are also many built-in ATH playbooks as well in the Stellar Cyber platform.

Number ten, Correlation with graph machine learning. This model analyzes the connections between identities, such as parent and child processes, and creates statistical scores to monitor for anomalies.

14:00-15:00

Cases within the Stellar Cyber platform utilizes Graph ML to visualize attacks and provide a detailed analysis and timeline. Correlation with Graph ML is possible with the Normalization and Enrichment and having access to all the different data sources. By being able to pull in consistent values, the platform can determine the timeline for a series of threats across all the different tools that exist within the organization and then identify the threat and map them together and show you how attacks progress. And also let you perform actions to mitigate the threat.

Number eleven, search. To enable the ability to locate data quickly and efficiently, data is stored as Normalized and Enriched Interflow records in the Stellar Cyber Data Lake.

15:00-16:00
All Interflow records are organized into indices, also known as indexes, based on the source of the data such as traffic, Alerts, assets, Linux events, Syslog, Windows events, and much more. Search and query tools in Stellar Cyber enable use of full text and regular expressions with Lucene query syntax to locate Interflow records. When field names and values within Interflow records are matched based on the search query the data may be displayed as a list of event records, all containing data specified in the search query. As an example, in the search tool you type in a query with Lucene query syntax. You can type in just a field name. You could type in just a field value, or you could do a combination of field names and values.

16:00-17:00
You can also use AND and OR and other operators as well, and you can use regular expressions in the query. The goal of typing the query is to locate records and display a list of events, or update charts, or even perform response actions based on an ATH playbook.

Number twelve, user interface. The user interface is a streamlined and intuitive web-based user interface for cybersecurity operations. It unifies and provides advanced visualization, Alerting, threat detection, automation, and response across the entire security landscape within an organization. So all the tools that we bring in from all the different sources are all able to be displayed, searched, and used within the Stellar Cyber platform user interface.

17:00-18:00
As an example, one of the visualizations you see is the XDR Kill Chain. The Kill Chain is just an example of some of the power of the user interface face. On the right side of the loop in red, you will see the Exfiltration events. If you click on that, that will perform a filter to show you the exfiltration events which are the most serious. On the left hand side of the loop at the beginning, are the initial attempts which are the least serious. If you click on that you can see all the events and charts related to that. There are many pages within the user interface and throughout the training you'll become more familiar with the Stellar Cyber user interface.

Number thirteen, Stellar Cyber application programming interface, otherwise known as the API. The Stellar Cyber API is an interface allowing programmatic interaction with the Stellar Cyber platform.

18:00-19:00
The Stellar Cyber API offers a range of functionalities including managing connectors, accessing detailed sensor and Case information, tenant creation, updating event tags and

statuses, performing queries, and much more. To use the Stellar Cyber API, people typically use a tool called Postman or write scripts to access the restful API. As an example, here's an API query to find 250 of the most recent cases reported by the Stellar Cyber platform. The API query contains, “https colon slash slash the IP address or host name of the Stellar Cyber instance slash connect slash API slash v one slash cases question mark and limit equals two fifty.” Again, this can all be adjusted.

19:00-20:00

There are many APIs that can be called. These are all documented in the Stellar Cyber platform documentation.

Number fourteen cloud connectors. Cloud connectors, or just connectors, run on the Stellar Cyber platform and utilize APIs to collect security and operational data from cloud security services, cloud infrastructure, and cloud apps This enables the ability to monitor, analyze, detect, and utilize automated and manual response actions to mitigate potential security threats. Cloud connectors, for example, allow you to collect data or create response actions for identity management solutions such as Azure, Okta, Google, email solutions such as Microsoft 365, Google Mail, EDR platforms to contain hosts or get data about or get threats...

20:00-21:00

...or Alerts from EDR platforms such as SentinelOne, CrowdStrike, and much more. You can get data from vulnerability, scanning services such as Nessus or Qualys. You can get information about cloud-based firewalls, CASB, WAF, NDR platforms such as Corelight and Darktrace. We even integrate with apps such as Azure AD, Salesforce, and ServiceNow as an example.

Number fifteen data sync. The data sync is a feature within the Stellar Cyber platform that enables the export of processed and enriched data to external systems or storage solutions. Data collected by the Stellar Cyber platform can simultaneously be stored in this Stellar Cyber Data Lake and exported to long-term storage solutions.

21:00-22:02

Other SIEMs and Data Lakes to facilitate broader data analysis and compliance with data-retention policies. Looking at the picture on step number fifteen, the data sync can simultaneously collect data and send data to AWS S3, Azure Blob, Google Cloud Storage, OCI archive, or other SIEMs like Splunk and even other Data Lakes such as Elasticsearch. Looking at the big picture on the right side, I have provided an overview of the Stellar Cyber Open XDR platform, the platform components and functionality, cloud connectors and data sync. Other modules will cover the rest of the big picture.

This lesson, *Open XDR Platform*, is complete. You should now be familiar with the major components and functionality of the Stellar Cyber Open XDR platform. Thank you.

L03 - STELLAR CYBER MODULAR SENSORS

L03A-PRES: Stellar Cyber Modular Sensors (11m:37s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Architecture Overview* module is *Modular Sensors.* This lesson, *Modular Sensors,* describes the components and features of Stellar Cyber Modular Sensors, which are essential for data collection from security, identity, infrastructure tools, and network traffic and for response actions to mitigate threats.

Looking at the big picture diagram for the Stellar Cyber Open XDR architecture, I will focus on the left side, which contains the Stellar Cyber Modular Sensor. This diagram gives an example of how a Stellar Cyber Modular Sensor connects to a switch within a network, which can be a physical network or virtualized network. The diagram shows some of the essential Modular Sensor features.

01:00-02:00
It shows how connectors on Modular Sensors connect to local security services and apps including Active Directory and firewalls and vulnerability scanners. It shows how logs are forwarded from infrastructures and security tools to the Stellar Cyber Modular Sensor. It shows the connections from the Modular Sensor to the Stellar Cyber Open XDR platform. I will start with number one: log forwarder and log filtering. Network security tools and infrastructure tools, including firewalls, routers, and switches, can forward their logs in the form of syslog including CEF and LEEF format, as well as JSON, NetFlow, and IPFIX, to Modular Sensors.

02:00-03:00
Modular Sensors use built-in parsers to interpret the logs and identify the format of the field names and field values within the logs. Logs from all sources are Normalized into consistent JSON format as Interflow records. Normalization is an essential process because all the different logs from all the different tools come in different formats, with different field names and field values. Stellar Cyber Modular Sensor uses the parsers to Normalize them as Interflow records. Modular Sensors then forward the logs as Interflow records to the Stellar Cyber platform. The Stellar Cyber platform can then use machine learning, statistical analysis rules, heuristic rules, custom ATH playbooks, and graph machine learning to identify security threats, generate Alerts, and correlate Alerts as Cases...

03:00-04:00
...that can be used by the SOC analysts to perform actions and to identify threats. When viewed as Cases, the SOC analyst can then decide how to process the Cases and all the Alerts within the Case. Response actions can also be performed by the SOC analyst or automatically to mitigate the threats. Log filters can be configured on Modular Sensors as well to optimize the data ingested by the Stellar Cyber platform by excluding specific log types that are not required for threat investigation or detection.

Number two: Network traffic analysis for network detection and response, also known as NDR. Network traffic from switches is sent to Modular Sensors via SPAN ports or TAP devices.

04:00-05:00
Looking at the picture on the left, you'll see that the Modular Sensor is connected to a switch, and a SPAN port is configured to forward traffic on the switch to the Modular Sensor. Modular Sensors use deep packet inspection, also known as DPI, to identify applications, create a session state, correlate app processes, and add metadata such as the number of bytes used by the application during a session. A summary of the traffic and metadata is forwarded as Interflow records to the Stellar Cyber platform. With the network traffic received, the Stellar Cyber platform uses machine learning, statistical analysis rules, heuristic rules, custom ATH playbooks, and graph machine learning to identify security threats, generate Alerts, and correlate Alerts as Cases. This might sound familiar and I hope it does.

05:00-06:00
Because network traffic analysis is one of the many ways that the Stellar Cyber Open XDR platform can ingest data to identify threats and enable the ability to perform response actions. Data filtering can also be configured to optimize the data ingested by this Stellar Cyber platform by including or excluding traffic and metadata from specific applications.

Number three: intrusion detection system, also known as IDS. After receiving network traffic from SPAN ports or TAP devices, Modular Sensors use a Suricata threat engine with sixty thousand plus commercial IDS rules to detect threats within network traffic, which also includes traffic from operational technology, also known as OT devices. The Modular Sensor also converts collected IDS data to metadata...

06:00-07:00
...then forwards the IDS metadata to the Stellar Cyber platform as Interflow records. After receiving the IDS metadata, the Stellar Cyber platform uses machine learning to reduce the number of events generated by the IDS engine and performs complex analysis and

correlation with other data sources to generate Alerts that may also be included within Cases.

Number four: Malware Sandbox. After receiving network traffic from SPAN ports or TAP devices, Modular Sensors identify, extract, and reconstruct malware traveling through the network within HTTP, which is simple web browsing, FTP which is the file transfer protocol. SMB, which is a file sharing protocol, and SMTP, which is a simple mail protocol. Modular Sensors forward files identified as malicious to an external HTTPS server for storage.

07:00-08:00
Modular sensors also forward malware metadata to the Stellar Cyber platform as Interflow records. From the Stellar Cyber platform, you can view and analyze observations of malware traveling through the network and arriving at specific hosts as well as view summaries of the events related to malware.

Number five: Aggregator. When configured as an Aggregator, the Modular Sensor acts as a proxy to forward traffic from other Modular Sensors, Windows Server sensors, and Linux Server sensors within the network to the Stellar Cyber platform. The Aggregator feature reduces the number of connections that are made from Modular Sensors and server sensors within an organization to the Stellar Cyber platform over the internet. When the Aggregator feature is not in use...

08:00-09:00
...all sensors forward Interflow records directly to the Stellar Cyber platform. With the Aggregator feature enabled, sensors can be configured to forward Interflow records to the Stellar Cyber platform through the Aggregator. The Aggregator feature is optional and can be used as a preference by the IT security administrators.

Number six: Tenable Nessus scanner. When enabled, a Tenable Nessus vulnerability scanner is installed on the Modular Sensor. The Tenable Nessus scanner instance will need to be linked to a valid Tenable.io account. After linking the scanner instance, you can use the Tenable.io site to initiate and monitor scans using the Tenable Nessus scanner on the Modular Sensor.

09:00-10:00
Number seven: Connectors. Connectors are software modules that when run on Modular Sensors can collect security and operational data from local security tools, identity providers, vulnerability scanners, and infrastructure devices. Connectors that run on Modular Sensors give the ability to monitor, analyze, detect threats, collect username information, and utilize automated and manual response actions to mitigate potential

security threats. Stellar Cyber has hundreds of connectors to infrastructure and security tools that may be used within an organization. This is one of the core components of being an Open XDR architecture.

Number eight: Buffering. With the log forwarding, network traffic metadata, and IDS features on Modular Sensors, buffering is enabled by default to temporarily store Interflow records locally.

10:00-11:00
If there is a connection issue to the Stellar Cyber platform, Modular Sensors temporarily hold the buffered Interflow records. When connections are restored to the Stellar Cyber platform, buffered Interflow records resume being forwarded at a trickle to prevent overwhelming the network connection. The amount of time logs are buffered depends on the configured buffer size. Looking at the big picture of the Stellar Cyber Open XDR architecture, I have covered the Stellar Cyber Modular Sensor, which is located on the left side of the diagram. I've covered the connectors, how logs are forwarded, how network traffic is sent to the Modular Sensor. I've covered how logs are forwarded from infrastructure tools to the Modular Sensor, how network traffic is used for network traffic analysis as well as IDS and Malware Sandbox.

11:00-11:37
I've covered how connectors can connect to local security services and apps. As well as other features including the Tenable Nessus scanner and buffering.

This lesson, *Modular Sensors*, is complete. You should now be familiar with the components and features of Stellar Cyber Modular Sensors, which are essential for data collection from security, identity, infrastructure tools, and network traffic, and for response actions to mitigate threats. Thank you.

L04 - STELLAR CYBER SERVER SENSORS

L04A-PRES: Stellar Cyber Server Sensors (04m:24s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Architecture Overview* module is S*erver Sensors*. This lesson, *Server Sensors*, provides an overview of how Windows Server sensors and Linux Server sensors collect and forward data to the Stellar Cyber Open XDR platform.

Looking at the big picture of the Stellar Cyber Open XDR architecture, I will focus on Stellar Cyber software that runs on Windows Servers, called Windows Server sensors and Stellar Cyber software that runs on Linux Servers, called Linux Server sensors. Looking at the big picture, the Windows Server sensors and Linux Server sensors both connect to the Stellar Cyber platform...

01:00-02:00

...with a configuration management connection, and there's a separate connection for where they send their Interflow records. Windows Server sensors and Linux Server sensors can also send connections to Modular Sensors set up as aggregators.

1. Windows Server sensors and Linux Server sensors are Stellar Cyber software installed on physical or virtual servers on premises or in the cloud to collect data used to detect threats on Windows Servers and Linux Servers. Once the data is collected, they send the data over a connection to the Stellar Cyber Open XDR platform. Windows Server sensors and Linux Server sensors can be configured from the Stellar Cyber Open XDR platform over the configuration management connection.

01:00-0:200

2. Windows Server sensors collect events within Windows Servers, from which they create Interflow records to send to the Stellar Cyber platform. Windows Server sensors collect the following information: They collect hardware events, security events, system events, Windows firewall events, Windows defender events, PowerShell events, and File Integrity Monitoring events, if it's enabled. Windows Server sensors Normalize and convert collected data to Interflow records and forward to the Stellar Cyber platform.
3. Linux Server sensors collect data within Linux Servers from which they create Interflow records to send to the Stellar Cyber platform. Linux Server sensors monitor and collect the following information.

03:00-04:00

They collect process information, command execution information about files, file events, and network traffic metadata. Linux Server sensors Normalize and convert collected data to Interflow records and forward to the Stellar Cyber platform. Once the data from Windows Server sensors and Linux Server sensors is collected by the Stellar Cyber Open XDR platform, the Stellar Cyber platform uses machine learning, statistical analysis rules, heuristic rules, ATH playbooks, and graph machine learning, to detect threats, correlate Alerts, and enable the ability to provide response actions to mitigate threats.

Looking at the big picture of the Stellar Cyber Open XDR architecture you can see now that the Windows Server sensors and Linux Server sensors are another essential component of the Stellar Cyber Open XDR architecture.

04:00-04:24
This lesson, *Server Sensors* is complete. You should now be familiar with how Windows Server sensors and Linux Server sensors collect and forward data to the Stellar Cyber Open XDR platform. Thank you.

L05 - DATA COLLECTION OVERVIEW

L05A-PRES: Data Collection Overview (07m:50s)

00:00-01:00
The next lesson in the *Stellar Cyber Open XDR Architecture Overview* module is *Data Collection Overview*. This lesson, *Data Collection Overview,* focuses on how the Stellar Cyber platform collects and ingests data from all the security, infrastructure, and identity tools within an organization to provide threat detection, investigation, and response actions to detect and mitigate threats.

I will start with the Stellar Cyber Open XDR platform and its major components. The Stellar Cyber Open XDR platform has a Data Lake. The Data Lake is the central repository for data storage in the Stellar Cyber platform. It aggregates and stores data from all sources, enabling the correlation of threats across all data inputs.

01:00-02:00
It also provides the foundation for data analysis and correlation. The Stellar Cyber user interface is a streamlined and intuitive web-based user interface for cybersecurity operations. It unifies and provides advanced visualization, Alerting, threat detection, automation, and response actions across the entire security landscape within an organization. The Stellar Cyber Open XDR platform also has an AI engine. It uses machine learning for threat detection and Alert creation. It also uses graph machine learning for correlation of Alerts into Cases. There's also a rules engine that includes statistical analysis rules, heuristic rules, and ATH playbooks all used for threat detection and Alert creation. As I go through how data is collected**…**

02:00-03:00

...it's important to understand two of the major components within the Stellar Cyber Open XDR architecture, which are the Stellar Cyber Modular Sensors and the Stellar Cyber Server sensors, including Windows Server sensors, and Linux Server sensors.

Number three. Organizations use cloud security tools and services. As an example, organizations use EDRs, which are endpoint detection and response solutions. Some examples of these are CrowdStrike, SentinelOne, and BlackBerry Cylance. Organizations also use email systems such as Microsoft 365 or Gmail. They also use CASB solutions, which are cloud access security broker solutions. An example of this is Netskope.

03:00-04:00
Organizations also use identity providers such as Microsoft Azure or Okta or Google. Organizations also use cloud apps such as Salesforce and cloud infrastructure such as AWS and Google Cloud and Oracle Cloud and cloud security services, such as AWS GuardDuty or CloudTrail.

Number four. The Stellar Cyber platform has hundreds of connectors that are used to connect to cloud security tools and services. These connectors collect data and enable response actions using these cloud security tools and services. For example, if a threat is detected, a response action can be to quarantine an endpoint using an EDR or sending an email or disabling a user account.

04:00-05:00
At number five, we can see that once data is collected from all the cloud security tools and services, the data is Normalized where all the field names and field values are put into a consistent JSON format as Interflow records, and they are Enriched with IP reputation from threat intel, geolocation of IP addresses, host names for IP addresses, usernames from identity providers, and IP type, such as public, private, or multicast IPs.

Number six, organizations also use on premises or cloud-hosted security tools and services such as Active Directory firewalls, network infrastructure including routers and switches. There's also network data that can be collected from on premises or cloud-hosted switches.

05:00-06:00
Other examples of on premises tools are vulnerability management tools.

At number seven, on premises and cloud-hosted security tools and services can send logs as Syslog and CEF or LEEF format, or JSON or IPFIX, or NetFlow to the Stellar Cyber Modular Sensor.

At number eight, the Stellar Cyber Modular Sensor Normalizes logs and metadata received from the network as JSON Interflow records and sends to the Stellar Cyber platform.

At number nine, the Interflow records are Enriched and stored within the Data Lake.

At number ten, connectors on Modular Sensors can connect to local services to obtain data and enable the ability to perform response actions such as disabling a user in Active Directory...

06:00-07:00
...or enable a firewall rule to block an IP address that is a threat. Once data is collected by the Stellar Cyber Modular Sensor, again it will Normalize the data and metadata as Interflow records.

At number twelve, the data is Enriched and stored as Interflow within the Data Lake.

At number thirteen Windows Server sensors and Linux Server sensors collect important data from Windows Servers and Linux Servers. That information is forwarded to the Stellar Cyber platform as Normalized Interflow records.

Those records at number fourteen are Enriched and stored in the Data Lake as Interflow. Once the Stellar Cyber Open XDR platform has collected all the data from all the different sources and has stored the Normalized and Enriched Interflow records within the Data Lake...

07:00-07:50
...the Stellar Cyber Open XDR platform AI engine and rules engine can be used to detect threats, create Alerts, and correlate all the threats that are from different sources into a unified view, making it much easier to understand and investigate threats as well as mitigate the threats by performing response actions to all the connected tools.

This lesson, *Data Collection Overview* is complete. You should now be familiar with how the Stellar Cyber platform collects and ingests data from all the security, infrastructure, and identity tools within an organization to provide threat detection, investigation, and response actions to detect and mitigate threats. Thank you.